Credit Risk Intelligence Platform

AI-Powered Loan Default Prediction & Business Intelligence

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EXECUTIVE DASHBOARD

Applicants Loaded: **307,511**
Default Rate: **8.07%**
Features Tracked: **719**
Risk Bands Configured: **LOW / MED / HIGH**
Test Set ROC-AUC: **0.7841**
Test Set Recall: **87.29%**

The Core Credit Risk Challenge & Solution

THE PROBLEM FOR BANKS

- **Extreme Class Imbalance:** Default rate is only 8.07%, making standard predictive algorithms bias heavily towards non-defaulters, leaving lenders vulnerable to bad loans.
- **Strict Regulatory Compliance:** Decisions must be explainable. model outputs (like gradient boosts) are black boxes that fail underwriting validation requirements.
- **Delayed Manual Underwriting:** Traditional scorecards are static. Reviewing loans manually takes hours or days, causing high client churn rates.
- **Inaccessible Data Silos:** Lenders cannot query live database metrics without SQL engineers, preventing business leaders from answering simple portfolio questions in real time.

THE CREDIT INTEL SOLUTION

- **Stacking Ensemble Classifier:** Combines Logistic Regression, Random Forest, and XGBoost with unsupervised Isolation Forest features to capture complex interactions while preserving calibration.
- **SHAP & LIME Local Explanations:** Provides dynamic local waterfall charts showing exactly which applicant-level attributes contributed to a risk decision.
- **Compliance Business Rules:** Generates clear IF/THEN rules extracted directly from the machine learning ensemble model using global surrogate decision trees.
- **Talk-to-Data SQL Assistant:** Allows credit analysts to query the database in plain English, generating and executing safe SELECT-only SQL scripts automatically.

System Architecture & Processing Pipeline

End-to-End Pipeline Stages

1. Data Loader & ETL
Loads Home Credit raw files, cleans outliers (such as DAYS_EMPLOYED placeholders), and outputs a Parquet feature store.

2. Preprocessor
Performs ColumnTransformer preprocessing: median imputation for missing numbers, mode imputation + OHE for categories, and standard scaling.

3. Isolation Forest
Calculates applicant-level anomaly scores which are appended directly to the feature matrix.

4. Stacking Ensemble
Trains Base Models (LR, RF, XGBoost) and feeds Out-of-Fold predictions to the Logistic Regression Meta-Learner.

5. Analytics & Serving
Persists model scoring logs to SQLite. Talk-to-Data intercepts and executes natural language inquiries via SQL translation.

Raw Datasets (CSV) ——> preprocessor.py ETL Loader

Feature Store Matrix (Parquet File format)

Isolation Forest (Appends Unsupervised Anomaly Score)

Stacking Models (LogReg + RandomForest + XGBoost)

Serving layer (FastAPI Backend + SQLite + Gemini)

Data Preprocessing & Feature Store Engineering

DATA PREPROCESSING PIPELINE

• **Deduplication & Cleaning:** Limits feature frame to 122 baseline columns; deletes duplicates. Translates negative variables like days of birth to positive metrics.

• **Outlier Correction:** Identifies anomalous entries. Replaces DAYS_EMPLOYED = 365,243 (which indicates unemployment in the Home Credit source) with NaN to prevent skewing.

• **Median Imputation:** Numerical missing columns are imputed with training set medians (e.g. EXT_SOURCE features).

•**One-Hot Encoding:** High-cardinality features are carefully evaluated and either grouped, encoded using alternative techniques, or excluded when they provide limited predictive value and significantly increase feature dimensionality.

• **Scaling & Splitting:** Applies Standard Scaling to all continuous features. Splits matrices into Train (60%), Validation (20%), and Test (20%) using stratified targets.

FEATURE ENGINEERING CATEGORIES

• **Credit Utilization Ratios:** Calculates AMT_CREDIT / AMT_INCOME_TOTAL and AMT_ANNUITY / AMT_INCOME_TOTAL to evaluate borrower leverage.

• **Payment Delay Features:** Derived from historical credit installments: STD and MEAN of late payment flags, max payment delay, and payment ratio sums.

• **Previous Loan Performance:** Calculates rejection rates of previous credit applications, credit-to-application ratios, and repayment speeds.

• **Bureau Aggregations:** Aggregates historical credit histories from credit bureau archives to calculate maximum debt-to-credit ratios.

• **Isolation Forest Anomaly Scores:** Fitted on preprocessed rows; computes multi-dimensional density score representing abnormality likelihood (appended as feature 719).

Stacking Ensemble & Classifier Implementation

STACKING METHODOLOGY RATIONALE

• **Diversity of Base Learners:** Combines three distinct machine learning approaches: Linear baselines (Logistic Regression), Bagging (Random Forest), and Boosting (XGBoost). This addresses different classification boundaries.

• **Calibration via Logistic Regression:** Base models generate probabilities. Stacking uses a Meta Logistic Regression classifier trained on Out-of-Fold (OOF) base model predictions. This corrects miscalibration in tree-based probabilities.

• **Class Imbalance Solutions:** Logistic Regression and Random Forest use balanced class weight penalties. XGBoost applies scale_pos_weight based on the negative/positive distribution (ratio ~ 11.4).

• **Feature Enrichment:** All models run on 719 features, including continuous demographics, bureau history aggregates, payment metrics, and anomaly scores.

BASE MODEL VALIDATION METRICS

Model Component	ROC-AUC	PR-AUC
Logistic Regression	0.7703	0.2453
Random Forest	0.7402	0.2140
XGBoost Classifier	0.7809	0.2691
Stacking Ensemble (Val)	**0.7798**	**0.2625**

Model Performance & Test Evaluation

TEST SET EVALUATION OUTCOMES

• **ROC-AUC Score: 0.7841:** Indicates strong discrimination ability. Defaulters score consistently higher than healthy accounts.

• **PR-AUC Score: 0.2830:** Reflects predictive reliability on highly imbalanced data (default baseline probability: 8.07%).

• **Recall: 87.29%:** At a custom underwriting threshold of 0.30, the model successfully identifies 87.29% of defaulting accounts.

• **Precision: 13.06%:** Represents the proportion of true default applicants within the flagged pool. Reflects the cost-accuracy trade-off for protecting capital.

• **Risk Band Thresholds:** Risk categorizations: LOW Risk (PD < 0.30), MEDIUM Risk (0.30 <= PD <= 0.60), and HIGH Risk (PD > 0.60).

TEST CONFUSION MATRIX (N = 61,503)

Actual / Predicted	Predicted Healthy (0)	Predicted Default (1)
Actual Healthy (0)	27,697 (True Neg)	28,841 (False Pos)
Actual Default (1)	631 (False Neg)	4,334 (True Pos)

Explainable AI (SHAP & LIME Integration)

MULTI-SCALE EXPLAINABILITY ENGINE

- **Global Feature Importance (SHAP):** Identifies overall drivers across the portfolio: EXT_SOURCE_2 (0.3274 Mean Abs SHAP) and EXT_SOURCE_3 (0.2666) represent external credit scores, while EXT_SOURCE_1 (0.1293) and CODE_GENDER (0.1093) are demographic/financial indicators.
- **Local SHAP Attribution:** Explains individual loan applications. Shows exactly how features push predicted defaults up or down relative to the baseline default probability (8.07%).
- **LIME Tabular Explanations:** Provides an alternate explainability engine. Approximates the complex stacking ensemble model locally with tabular linear models to confirm SHAP output.
- **Live Waterfall Plots:** Generates real-time waterfall graphics showing the path from baseline average defaults to the final prediction score.

TOP PORTFOLIO RISK DRIVERS (SHAP)

Feature Name	Mean Absolute SHAP Value
EXT_SOURCE_2	0.3274
EXT_SOURCE_3	0.2666
EXT_SOURCE_1	0.1292
CODE_GENDER	0.1093
APP_GOODS_TO_CREDIT_RATIO	0.1032

Business Rules Engine (Decision Tree Surrogate)

COMPLIANCE & UNDERWRITING RULES

- **Global Decision Tree Surrogate:** We train a Decision Tree Classifier (max_depth=4, min_samples_leaf=200) to approximate the complex stacking ensemble. This makes the predictions auditable.
- **Auto Rule Extraction:** Automatically translates decision splits from the surrogate tree into logical boolean statements for credit officers.
- **Specific Customer Rules:** Combines local SHAP drivers and risk bands to generate specific applicant rules, explaining exactly why an individual was categorized as High or Medium risk.
- **Rule Audit Output:** All rules are logged as JSON and markdown files under documents/rules/ for automated compliance review.

EXTRACTED COMPLIANCE RULES

RULE PATH 1 (SURROGATE TREE):
IF EXT_SOURCE_3 <= -0.864
AND EXT_SOURCE_2 <= -0.035
AND EXT_SOURCE_1 <= 0.312
AND BUREAU_DEBT_TO_CREDIT_RATIO <= -0.007
THEN HIGH RISK

RULE PATH 2 (SURROGATE TREE):
IF EXT_SOURCE_3 > -0.864
AND EXT_SOURCE_2 <= -0.593
AND EXT_SOURCE_3 <= 0.168
AND EMPLOYMENT_YEARS <= -0.283
THEN HIGH RISK

Talk-to-Data Natural Language Analytics Engine

TALK-TO-DATA ASSISTANT WORKFLOW

- **Double-Layer Translation Engine:** Uses Gemini 2.5 API with specific system instructions and schema specifications. Safely translates plain English to SELECT-only SQL queries.
- **AST SQL Validation Security:** An Abstract Syntax Tree (AST) SQLValidator checks generated queries. Blocks destructive keywords (e.g. DELETE, DROP, INSERT) and enforces a LIMIT 100 on output records.
- **Double-Connection Database Isolation:** Fires queries against SQLite database via a read-only connection, protecting sensitive database records.
- **Insight Generator Layer:** Summarizes the returned rows from SQLite using an analyst persona template to output business insights directly to the user.

TALK-TO-DATA DIALOGUE EXAMPLE

Credit Analyst: "Which occupation experiences the highest default rate?"

Generated SQL Query:

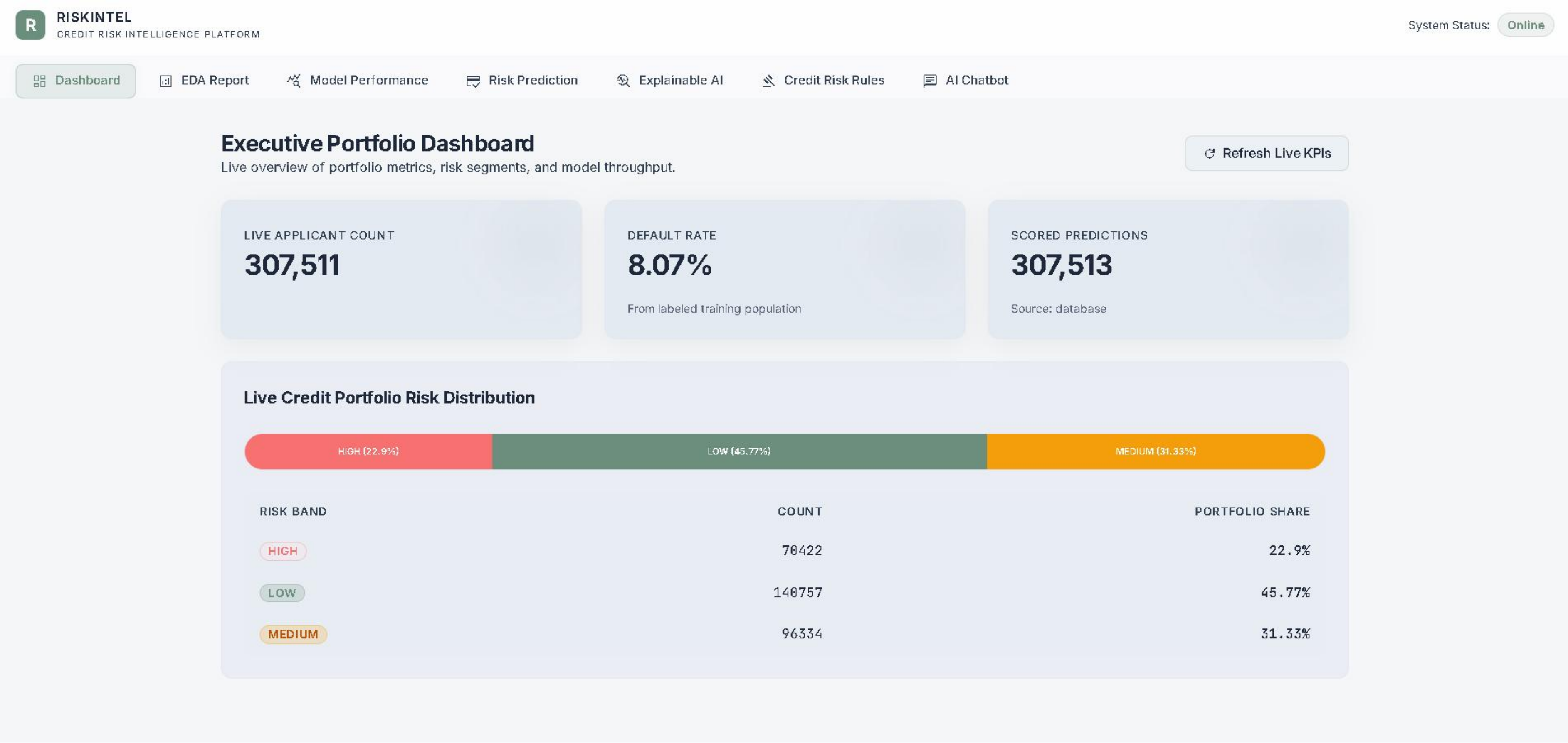
```
SELECT occupation_type, AVG(target) AS default_rate
FROM customers GROUP BY occupation_type
ORDER BY default_rate DESC LIMIT 1
```

Assistant Insight Response:
"Low-skill laborers experience the highest default rates in the portfolio at 17.15%, which is over twice the average baseline rate of 8.07%. This indicates a need for tighter credit limits on applicants in this segment."

Application User Interface & Screens

1. Executive Dashboard

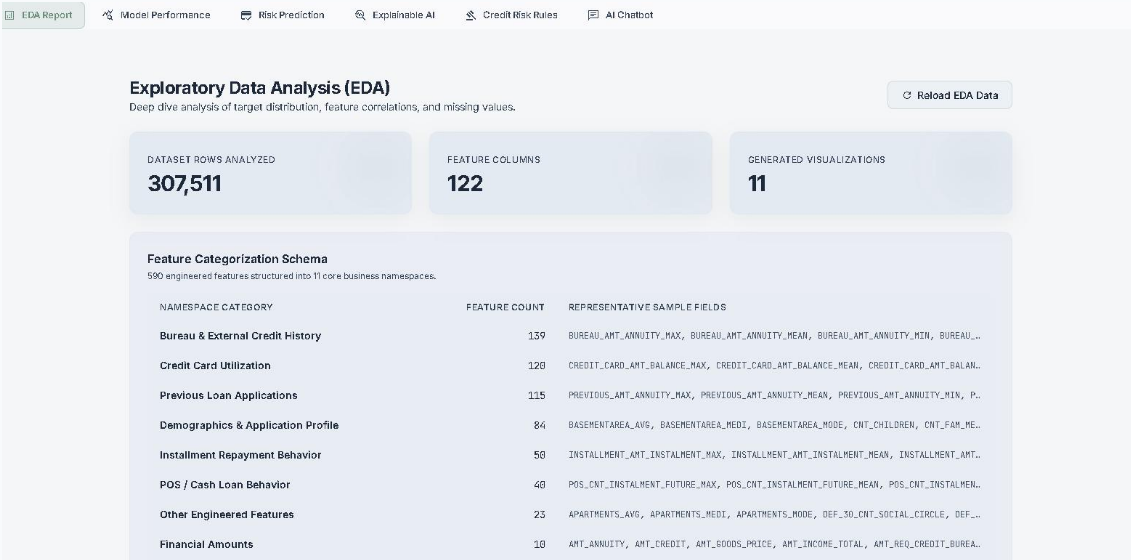
Provides a high-level overview of the credit portfolio and model activity. Displays key business metrics such as Total Applicants, Portfolio Default Rate, Risk Distribution, Prediction Logs, and Risk Category Breakdown. Interactive visualizations help stakeholders quickly assess portfolio health and lending risk trends.

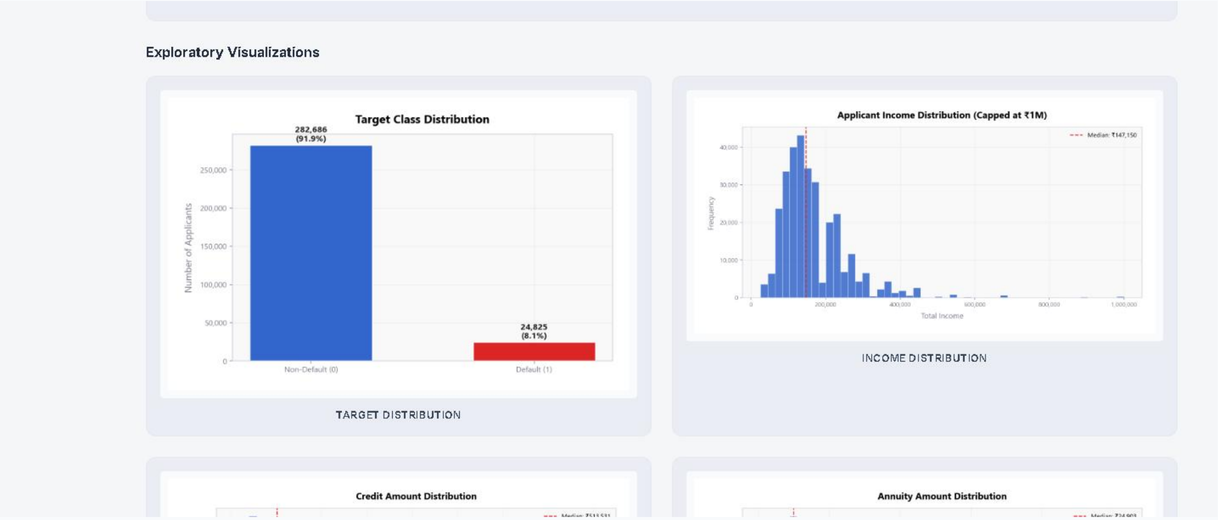


Application User Interface & Screens

2. EDA & Analytics Portal

Dedicated data exploration interface used for understanding customer demographics, financial behavior, and credit characteristics. Includes distributions, correlation analysis, missing value analysis, repayment behavior trends, and risk segmentation charts that support data-driven decision making.



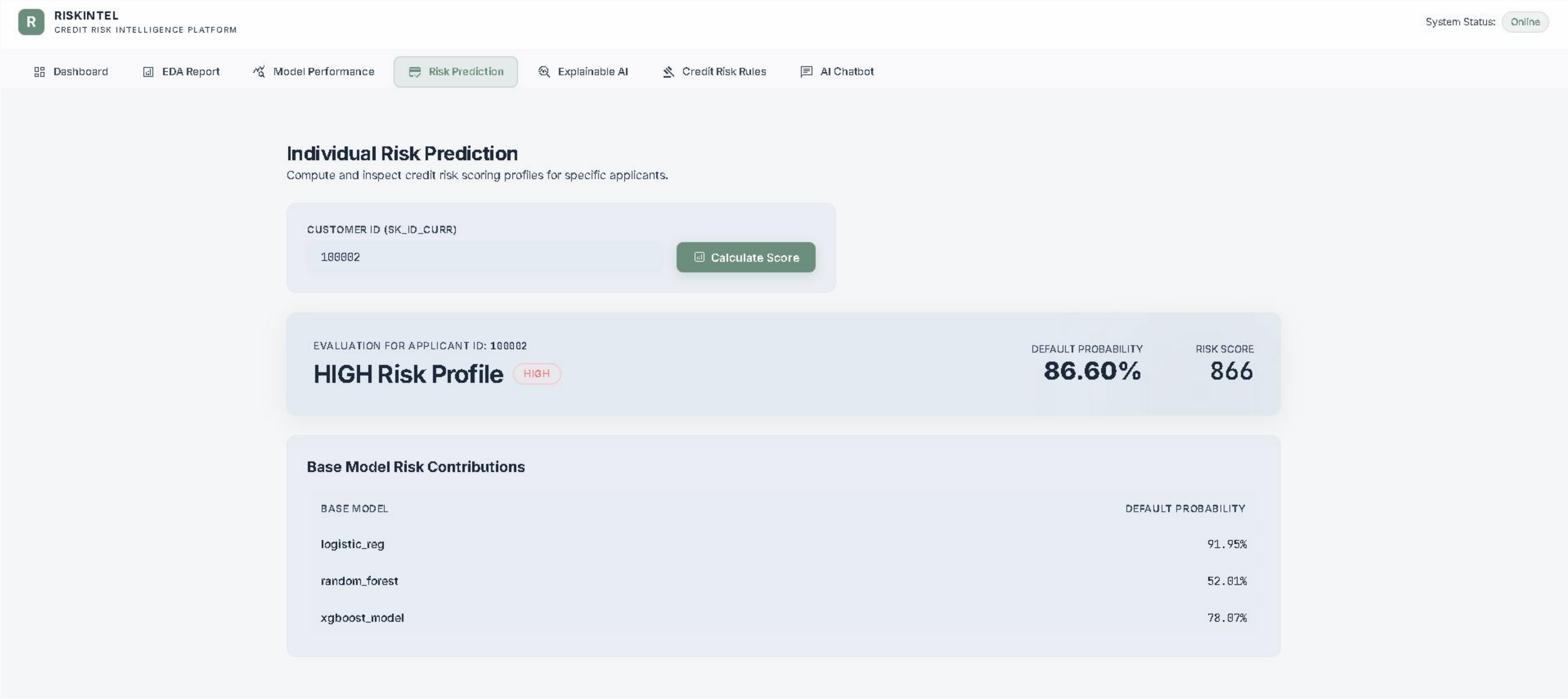


CREDIT RISK INTELLIGENCE PLATFORM

Application User Interface & Screens

3. Loan Default Prediction Page

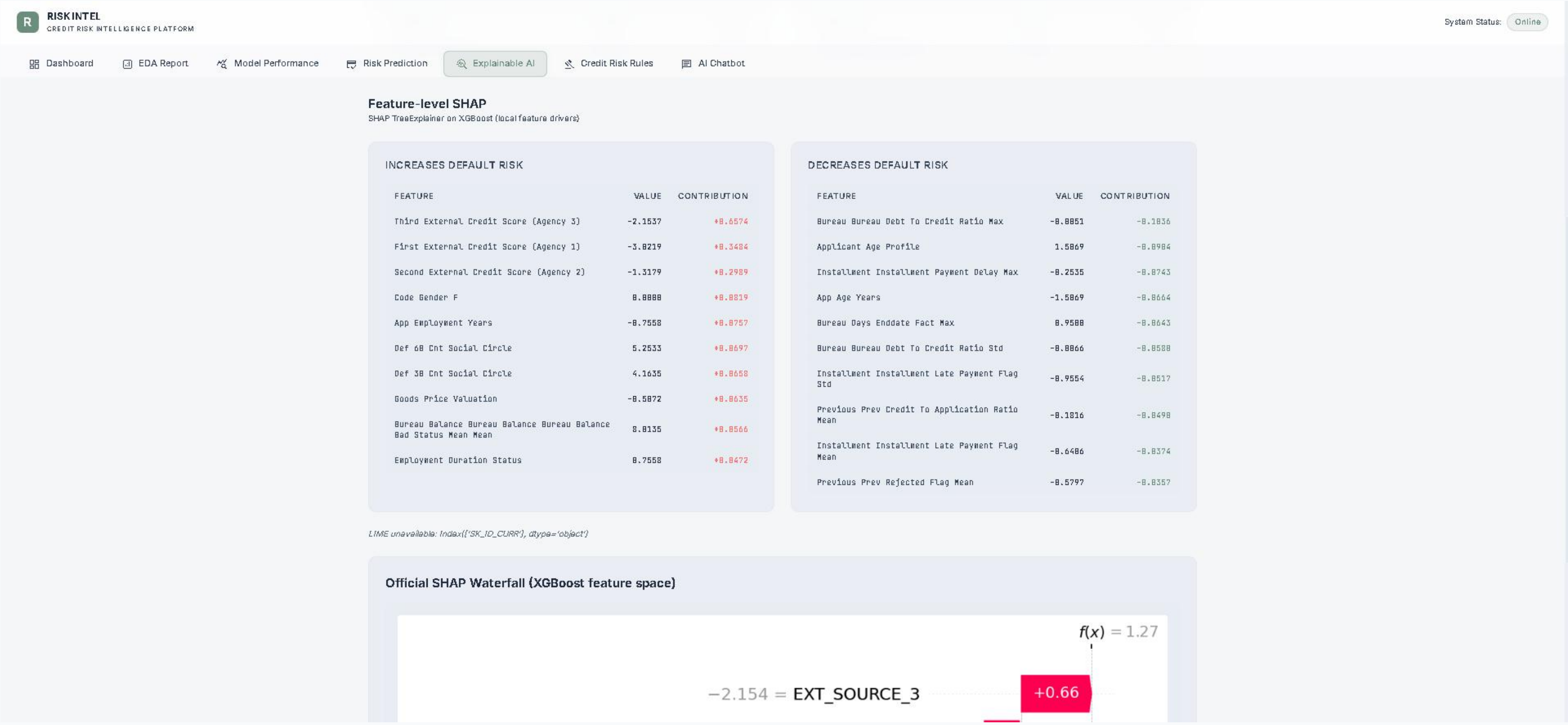
Enables real-time credit risk assessment. The system outputs default probability, risk score, and LOW/MEDIUM/HIGH risk classification, supporting faster underwriting decisions.



Application User Interface & Screens

4. Explainability & Decision Intelligence Portal

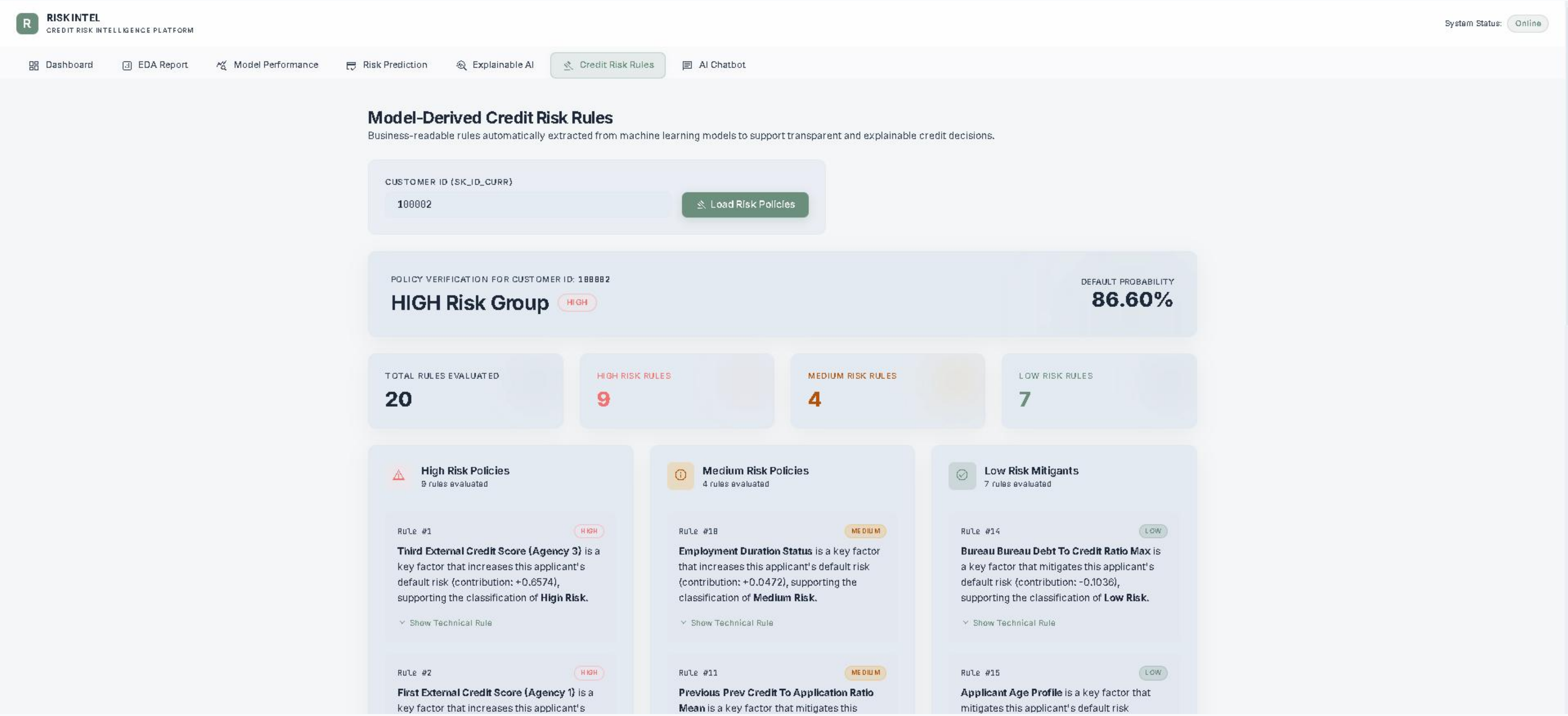
Provides transparency into machine learning predictions using Explainable AI techniques. Displays SHAP-based feature contribution analysis, waterfall charts, and localized decision explanations. Surrogate decision tree rules are also shown to help business users understand why an applicant was classified as risky.



Application User Interface & Screens

5. Business Rules & Policy Explorer

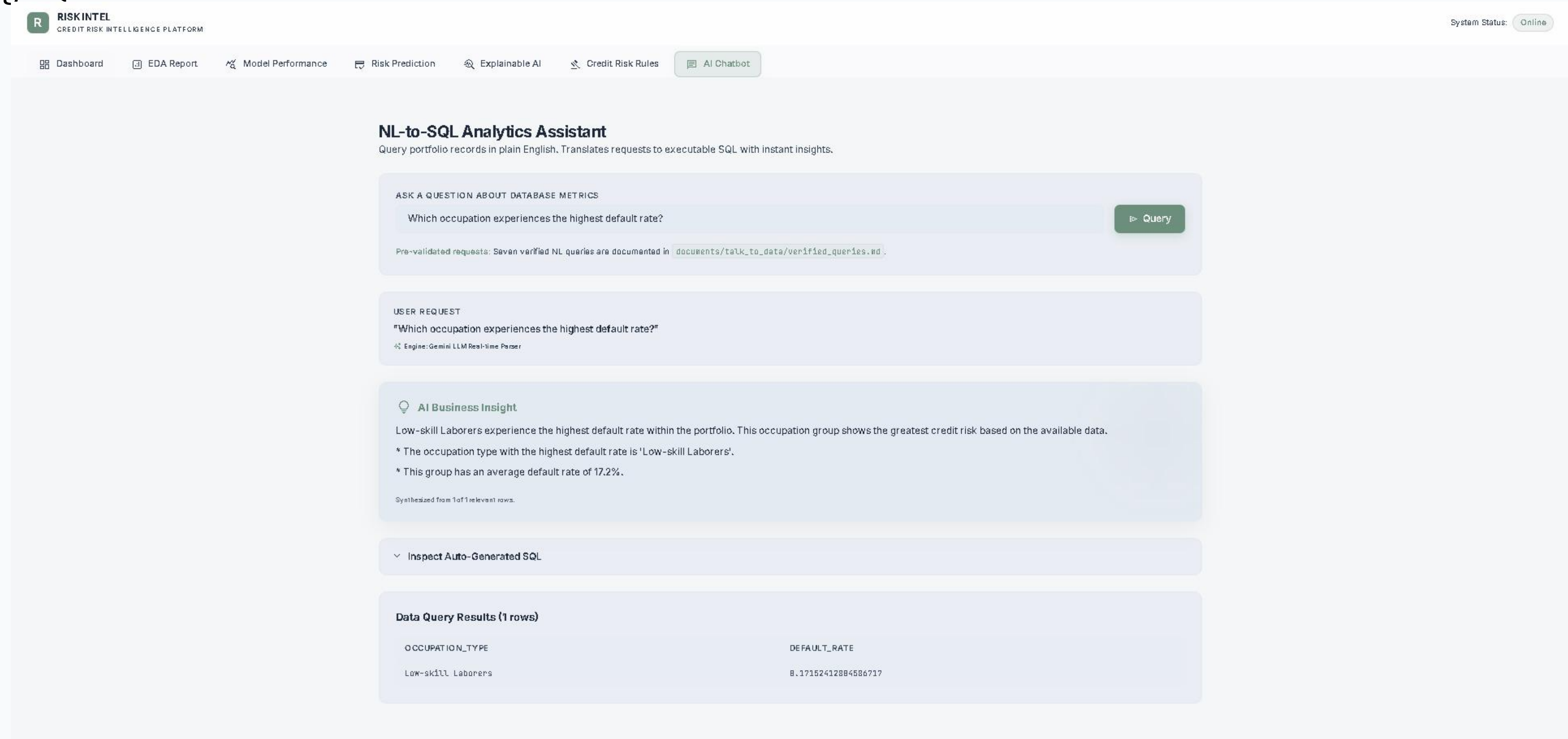
Converts machine learning insights into business-readable decision rules. Users can browse generated policies, explore risk conditions, and understand model behavior through interpretable IF–THEN rules that support compliance, auditability, and credit policy development.



Application User Interface & Screens

6. Talk-to-Data Interactive Chatbot

A conversational analytics interface powered by Natural Language to SQL technology. Users can ask business questions in plain English, view the generated SQL query, inspect returned data records, and receive AI-generated business summaries for rapid portfolio analysis without writing SQL.



System Engineering & Containerized Deployment

ENGINEERING PLATFORM HIGHLIGHTS

✓ **Modular Architecture Separation:** Keeps pipeline loader code, modeling modules, API endpoints, SQL validators, and frontend structures strictly decoupled.

✓ **Pydantic Environment Configs:** All settings (thresholds, model paths, model variables) are managed using clean Pydantic Settings connected to .env configuration files.

✓ **Rigorous Error Handling & Logging:** API endpoints catch and log database validation, preprocessing, and scoring exceptions to file-based logs.

✓ **SQLast Execution Shield:** Validates translated SQL with abstract syntax structures to block malicious queries, securing SQLite database tables.

ONE-STEP DOCKER DEPLOYMENT

1. Setup Environment Variables:

```
cp .env.example .env
nano .env  # Add your GEMINI_API_KEY
```

2. Build & Launch Containers:

```
docker compose up --build
```

Auto-Training Logic:
If base model files or meta-learners are missing in the /models directory, Docker Compose triggers model training automatically on startup before booting FastAPI.